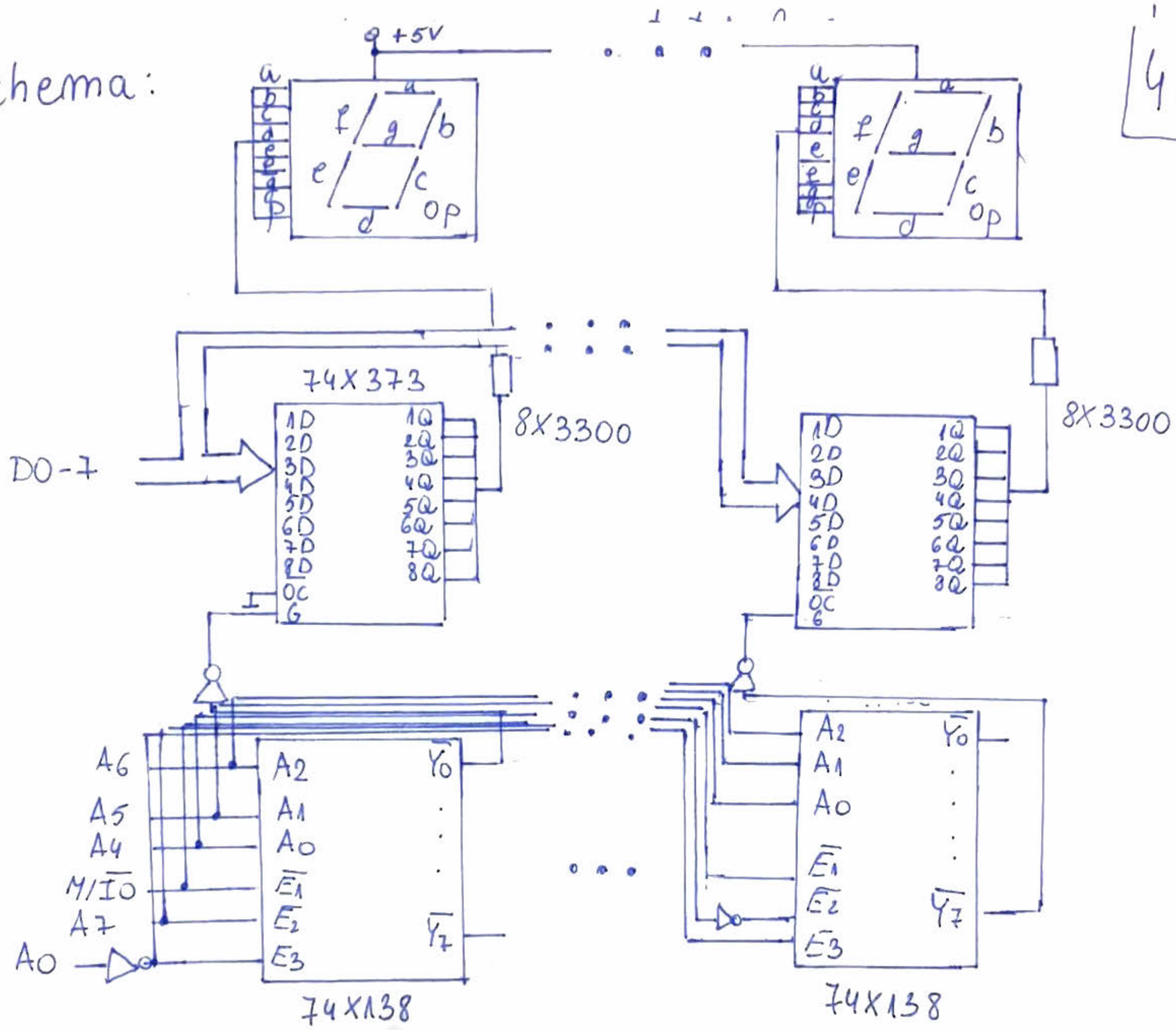
(10.) Să se conecteze la o unitate centrală cu microprocesorul 8086, un modul de afișare cu 16 segmente. Se va scrie câte o rutină pentru afișarea câte unei cifre hexazecimale pe fiecare rang.

Rezolvare:

schema:


+5V
a
b
c
d
e
f
g
p
74X373
1D
2D
3D
4D
5D
6D
7D
8D
OC
G
1Q
2Q
3Q
4Q
5Q
6Q
7Q
8Q
D0-7
8X3300
A6
A5
A4
M/IO
A7
A0
A2
A1
A0
E1
E2
E3
Y0
Y7
74X138

Programul

; afișare cifrei 0 => f | a | b, e | d | c (rang 1)

p	g	f	e	d	c	b	a
1	1	0	0	0	0	0	0

=> C0H

```
MOV AL, 0C0H
OUT 00H, AL
```

; afișarea cifrei 1 => b, c (rang 2)

p	g	f	e	d	c	b	a
1	1	1	1	1	0	0	1

=> F9H

```
MOV AL, 0F9H
OUT 10H, AL
```

; afișarea cifrei 2 => a, g, b, e, d (rang 3)

p	g	f	e	d	c	b	a
1	0	1	0	0	1	0	0

=> A4H

```
MOV AL, 0A4H
OUT 20H, AL
```

; afișarea cifrei 3 => a, g, b, c, d (rang 4)

p	g	f	e	d	c	b	a
1	0	1	1	0	0	0	0

=> B0H

```
MOV AL, 0B0H
OUT 30H, AL
```

; afișarea cifrei 4 => f, g, b, c (rang 5)

p	g	f	e	d	c	b	a
1	0	0	1	1	0	0	1

=> 99H

```
MOV AL, 099H
OUT 40H, AL
```

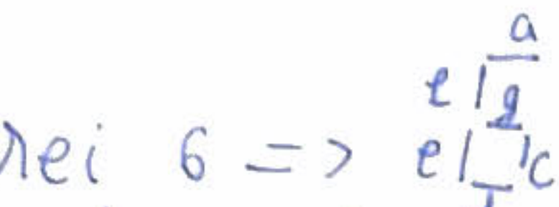
afișarea cifrei 5 =>

(rang 6)

p	g	f	e	d	c	b	a
1	0	0	1	0	0	1	0

=> 92H

```
MOV AL, 92H
OUT 50H, AL
```

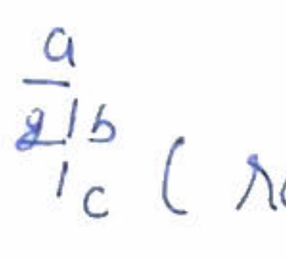
afișarea cifrei 6 => 

(rang 7)

p	g	f	e	d	c	b	a
1	0	0	0	0	0	1	0

=> 82H

```
MOV AL, 82H
OUT 60H, AL
```

afișarea cifrei 7 => 

(rang 8)

p	g	f	e	d	c	b	a
1	0	1	1	1	0	0	0

=> B8H

```
MOV AL, B8H
OUT 70H, AL
```

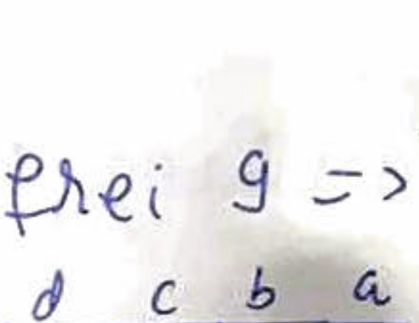
afișarea cifrei 8 =>

(rang 9)

p	g	f	e	d	c	b	a
1	0	0	0	0	0	0	0

=> 80H

```
MOV AL, 80H
OUT 80H, AL
```

afișarea cifrei 9 => 

(rang 10)

p	g	f	e	d	c	b	a
1	0	0	1	0	0	0	0

=> 110H

```
MOV AL, 110H
OUT 90H, AL
```

; afișarea literei A => f|a|b, e|g|c (rang 11)

p	g	f	e	d	c	b	a
1	0	0	0	1	0	0	0

=> 88H

```
MOV AL, 88H
OUT A0H, AL
```

; afișarea literei B => f|g, e|_|c, d (rang 12)

p	g	f	e	d	c	b	a
1	0	0	0	0	0	1	1

=> 83H

```
MOV AL, 83H
OUT B0H, AL
```

; afișarea literei C => f|a, e|_, d (rang 13)

p	g	f	e	d	c	b	a
1	1	0	0	0	1	1	0

=> C6H

```
MOV AL,0C6H
OUT C0H, AL
```

; afișarea literei D => g|b, e|_|c, d (rang 14)

p	g	f	e	d	c	b	a
1	0	1	0	0	0	0	1

=> A1H

```
MOV AL, 0A1H
OUT D0H, AL
```

; afișarea literei E => f|a, e|_|g, d (rang 15)

p	g	f	e	d	c	b	a
1	0	0	0	0	1	1	0

=> 86H

```
MOV AL, 86H
OUT E0H, AL
```

și afișarea literei F => $\frac{e}{e1}\frac{a}{g}$ (rang 16)

p	g	f	e	d	c	b	a
1	0	0	0	1	1	1	0

=> 8EH

```
MOV AL, 8EH
OUT F0H, AL
```